

Supplementary Information

Cross-family disruption of monoamine transporter assembly by a NET-derived truncation isoform revealed through near-million-atom simulations

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Table S1. Computationally mapped and predicted potential truncated isoforms of the monoamine transporter (MAT) family (NET, DAT, SERT).

Name	Isoform ID ¹	TM count ²	Length (aa) ³	Mass (Da) ⁴	Isoelectric point (pI) ⁵	GRAVY ⁶ (cut-off=0)
NET	P23975 (Canonical)	12	617	69332	7.18	0.468
	P23975-2	12	628	70548	8.44	0.414
	P23975-3	10	512	57370	6.26	0.568
	A0A804HLI4	6	350	39071	8.01	0.175
	H3BRS0	5	295	33164	8.57	0.222
DAT	Q01959 (Canonical)	12	620	68495	6.46	0.499
SERT	P31645 (Canonical)	12	630	70325	5.89	0.422
	P31645-2	12	672	74978	5.91	0.336
	J3QKP3	1	72	7889	5.56	1.149

¹The Uniprot entry ID of the isoform (only included isoforms that have length between 10% and 90% of the canonical sequence), P23975-2 and P31645-2 have also been added for comparison purposes. Please see Methods.

²Transmembrane (TM) domain count of the protein, derived from the topology information included in the Uniprot entries and Protter.

³The amino acid length of the isoform sequence.

⁴Protein mass calculated from sequences via ProtParam.

⁴Protein isoelectric point calculated from sequences via ProtParam.

⁵The GRAVY (Grand Average of Hydropathy) value for the corresponding protein is calculated as the sum of hydropathy values of all the amino acids, divided by the number of residues in the sequence.

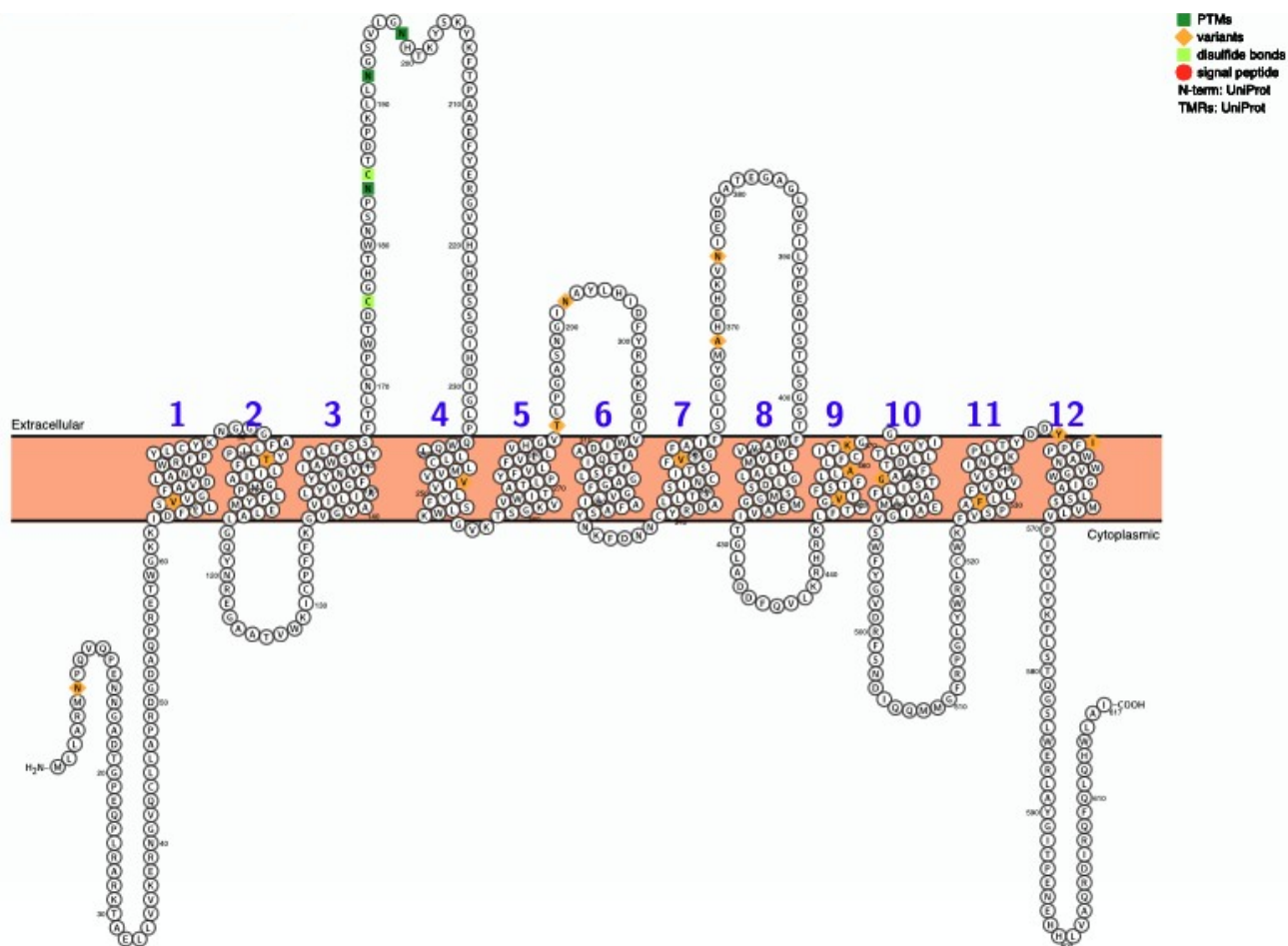


Figure S1. Sequence and topology visualization of the canonical NET isoform P23975 from Protter.

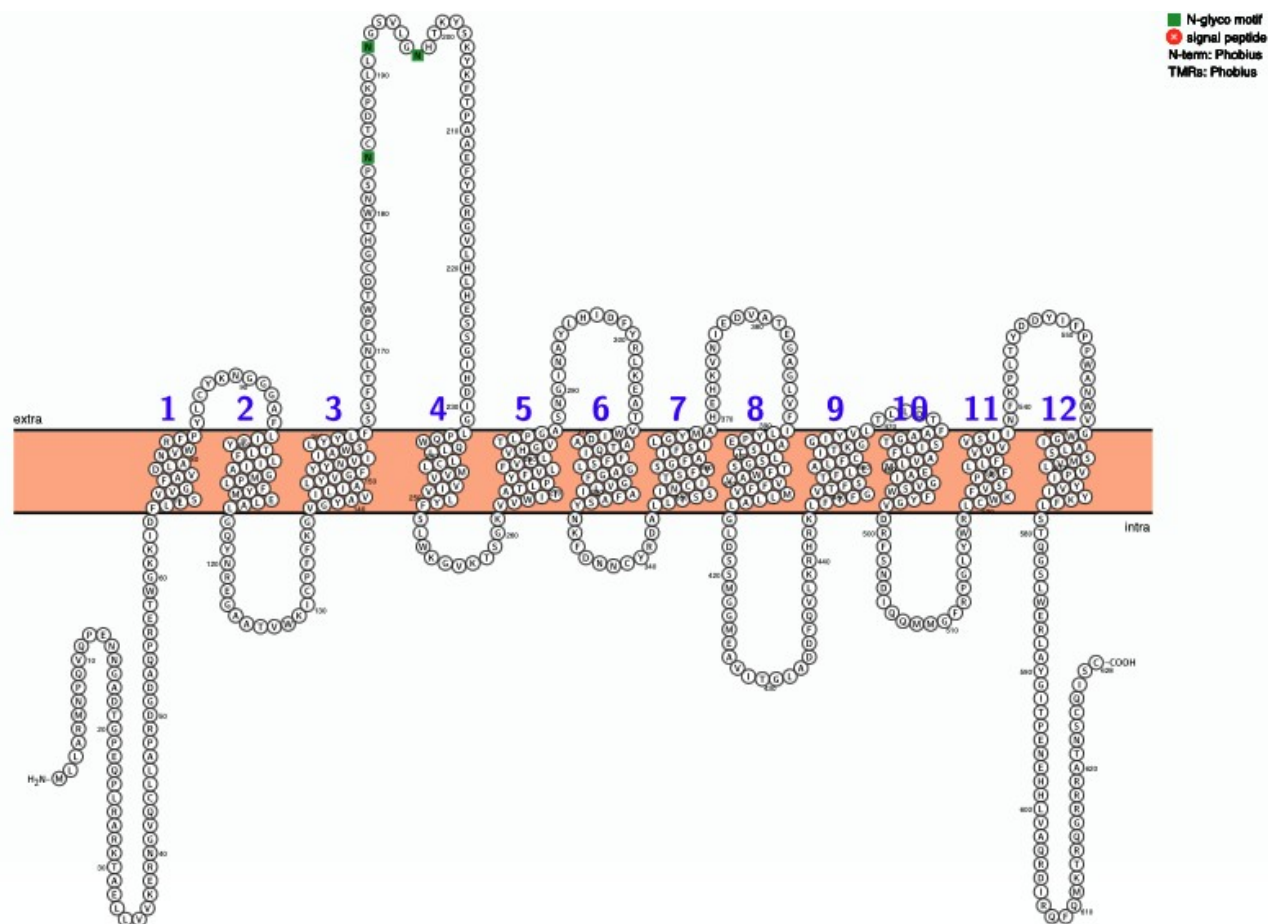


Figure S2. Sequence and topology visualization of the NET isoform P23975-2 from Protter.

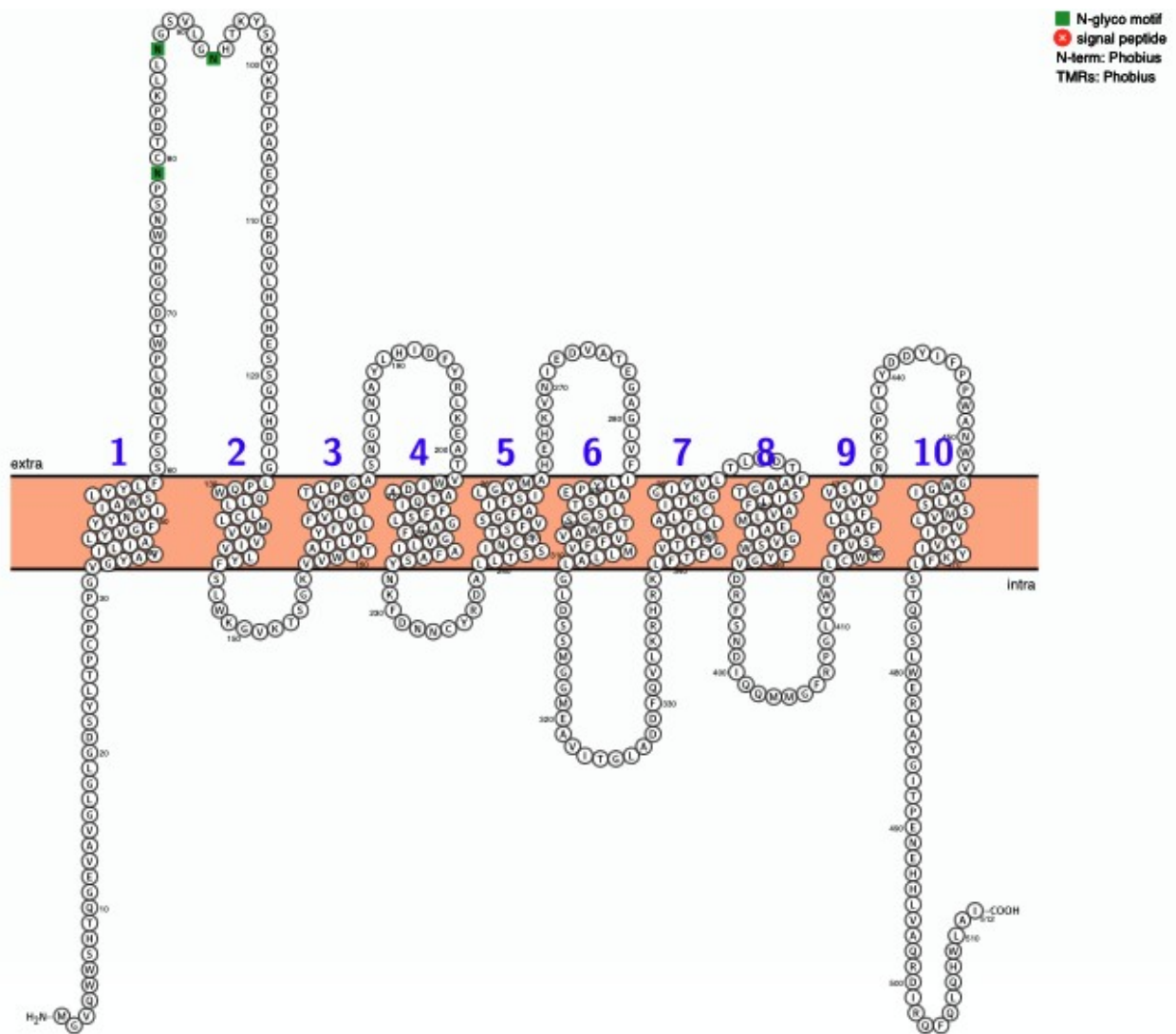


Figure S3. Sequence and topology visualization of the NET isoform P23975-3 from Protter.

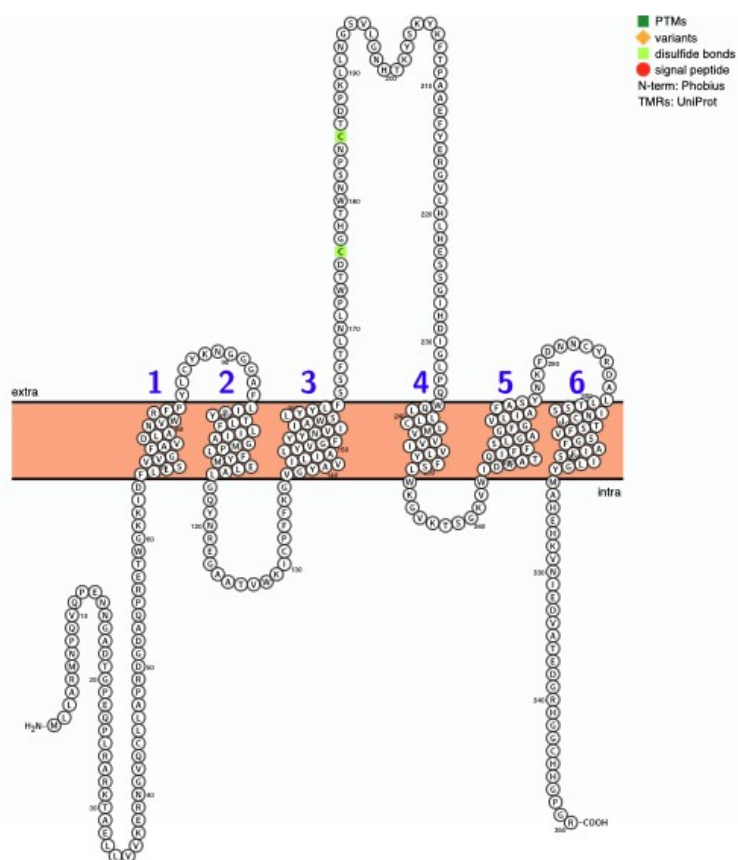


Figure S4. Sequence and topology visualization of the NET isoform A0A804HLI4 from Protter.

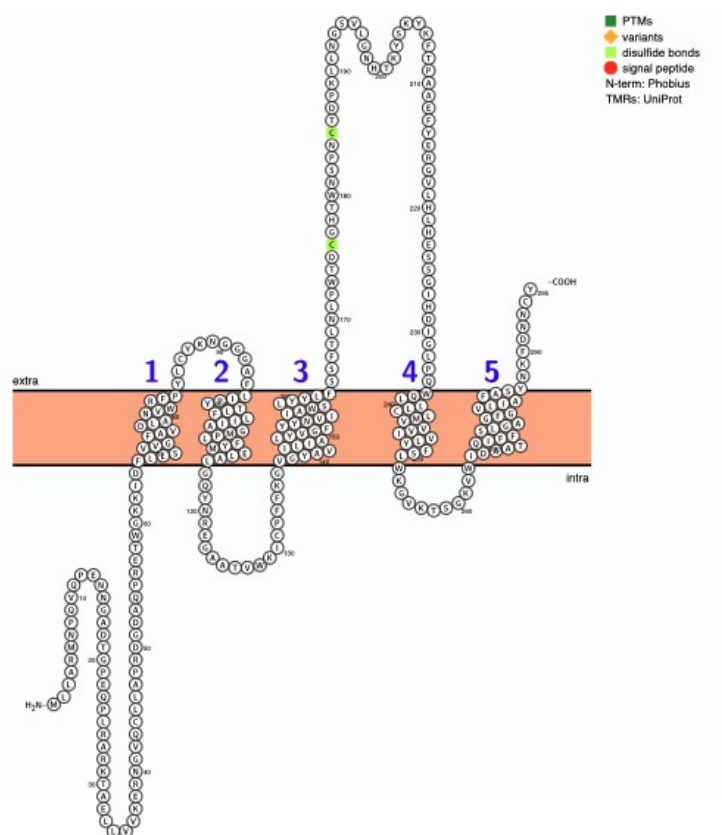


Figure S5. Sequence and topology visualization of the NET isoform H3BRS0 from Protter.

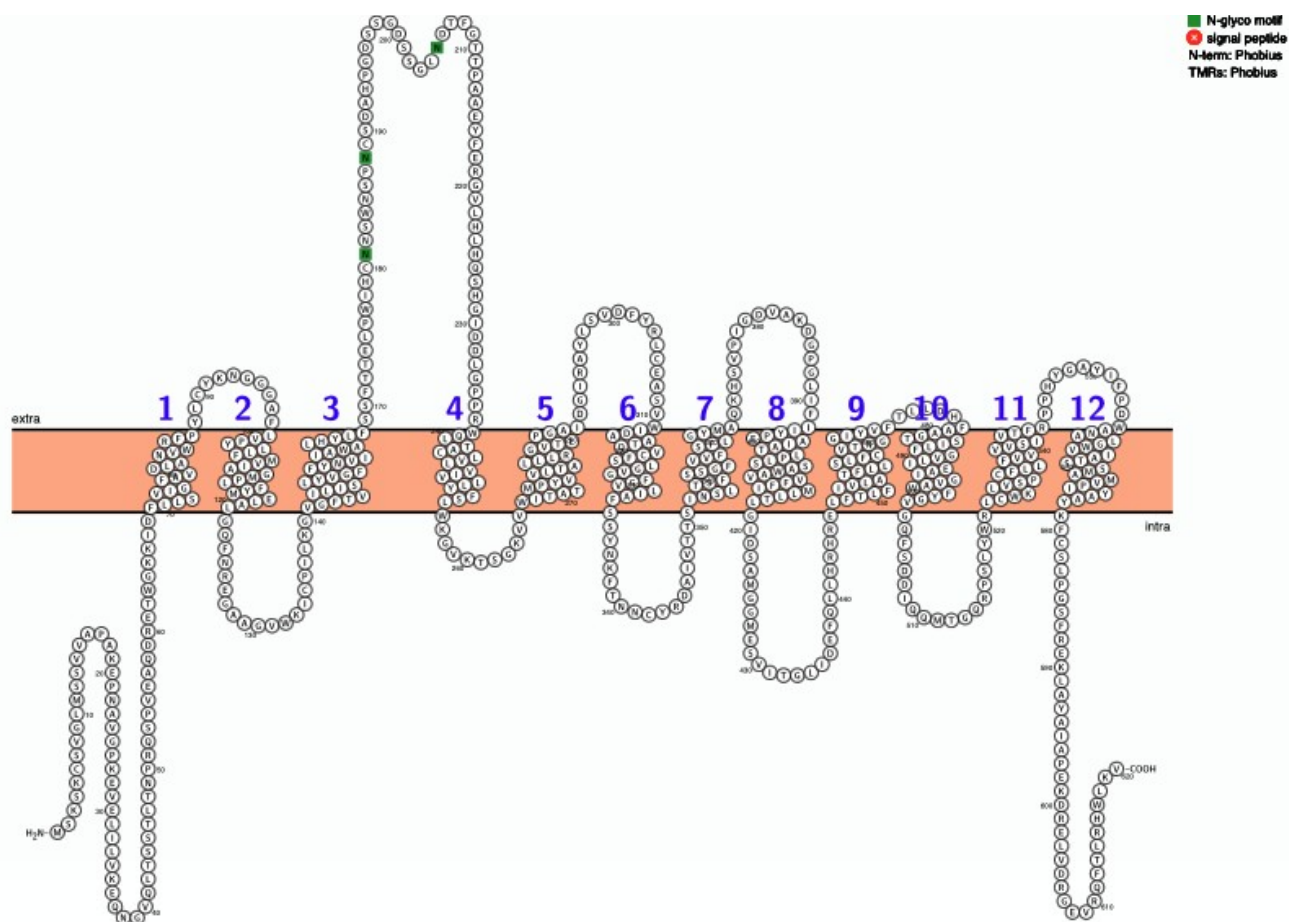


Figure S6. Sequence and topology visualization of the canonical DAT isoform Q01959 from Protter.

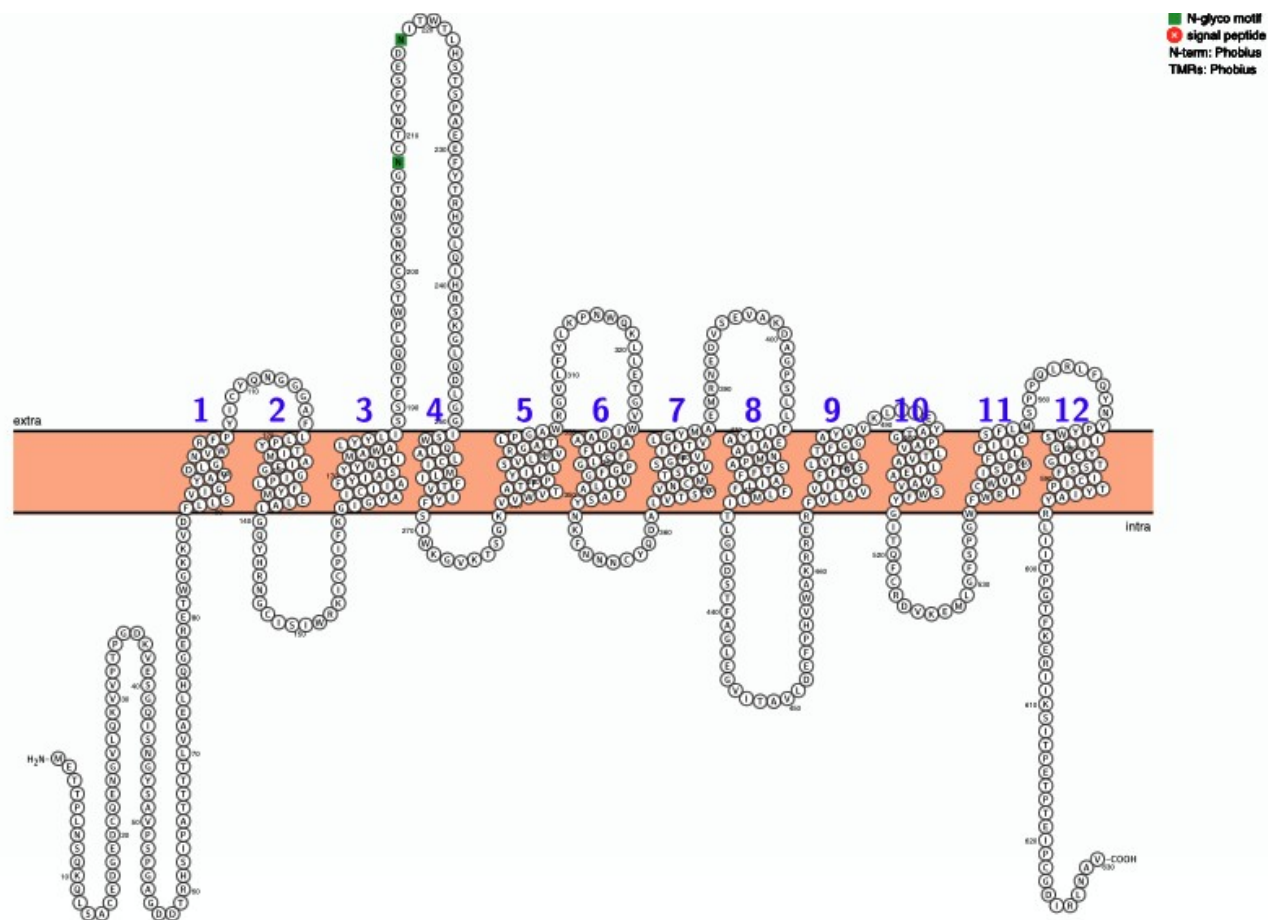


Figure S7. Sequence and topology visualization of the canonical SERT isoform P31645 from Protter.

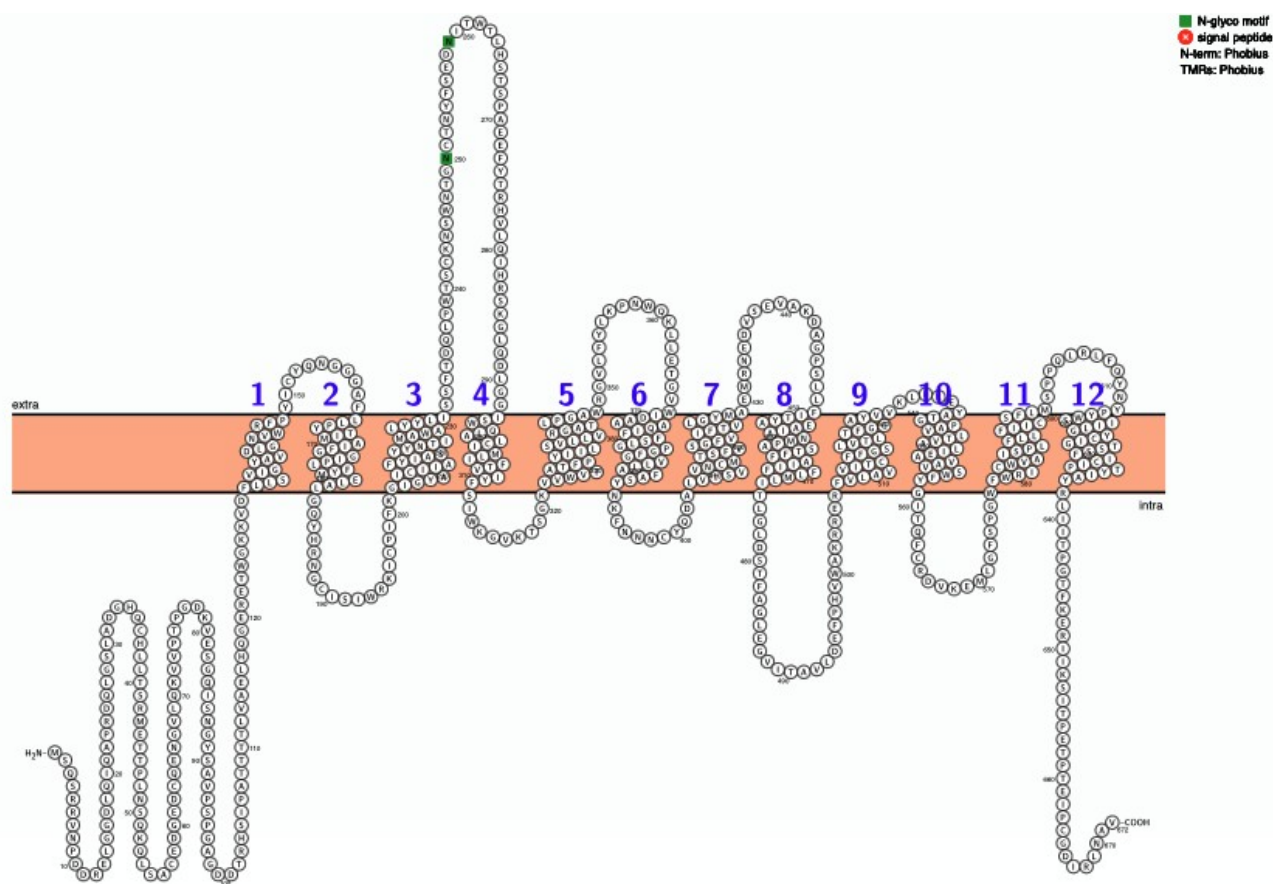


Figure S8. Sequence and topology visualization of the SERT isoform P31645-2 from Protter.

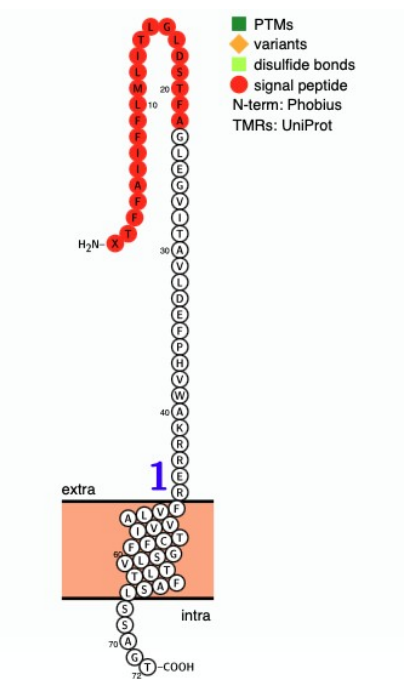


Figure S9. Sequence and topology visualization of the SERT isoform J3QKP3 from Protter.

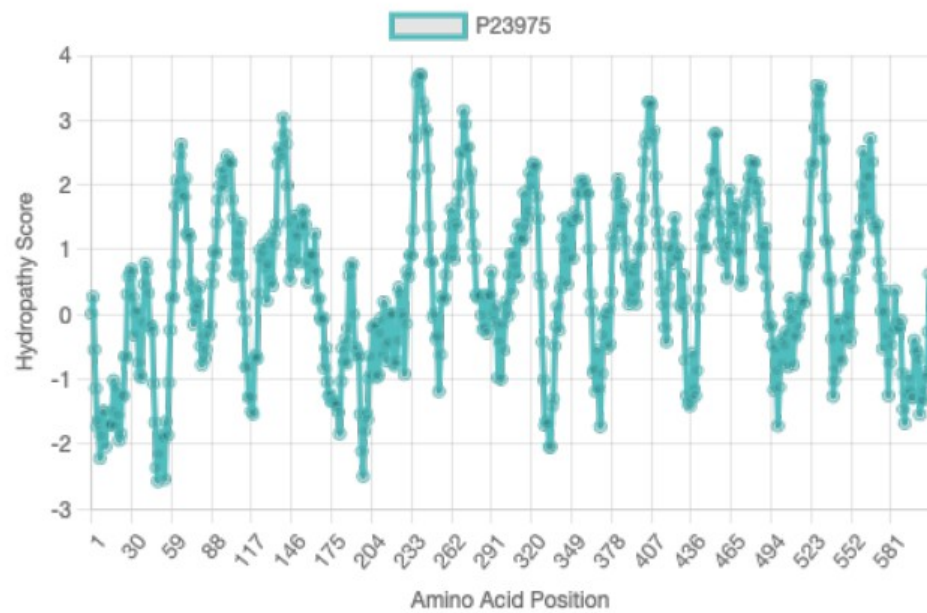


Figure S10. Kyte-Doolittle hydropathy plot of the canonical NET isoform P23975.

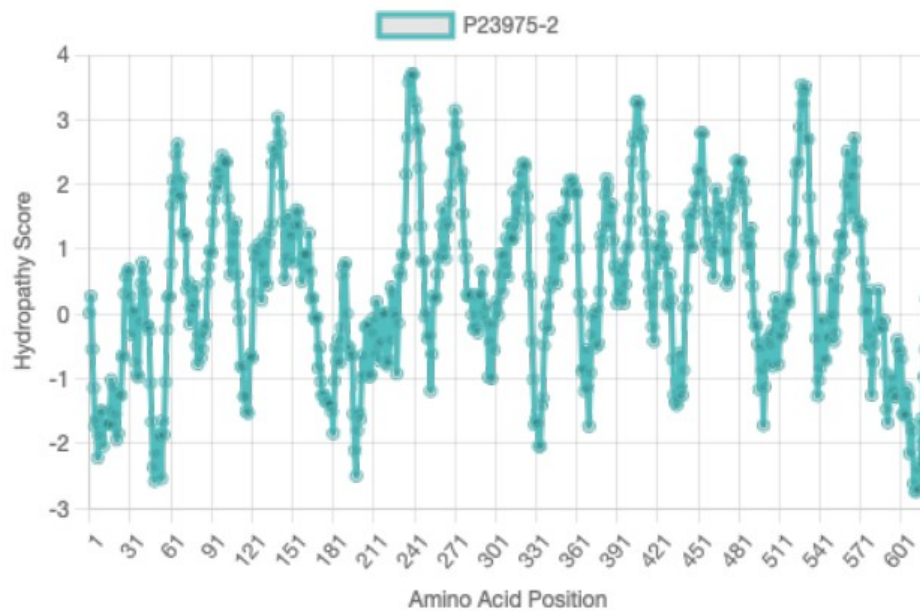


Figure S11. Kyte-Doolittle hydropathy plot of the NET isoform P23975-2.

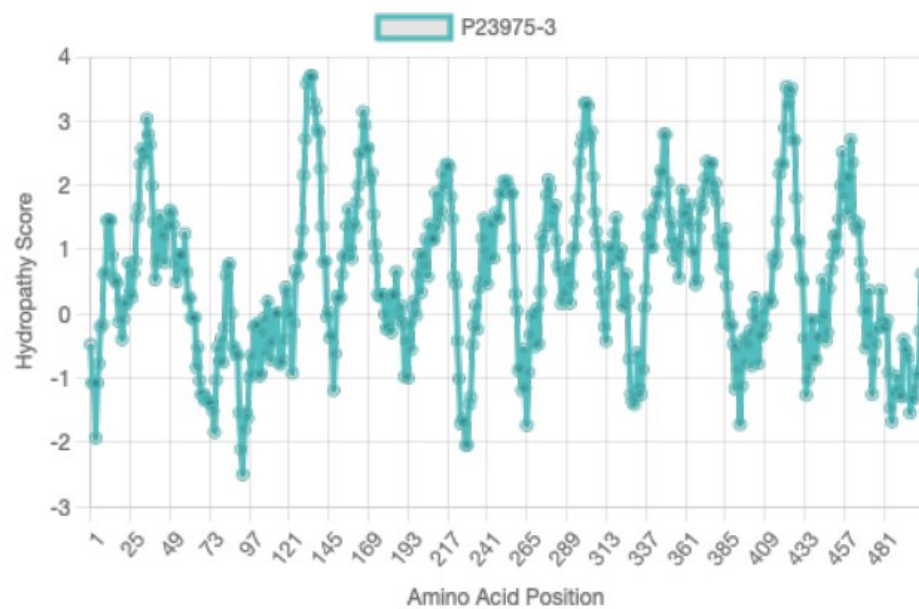


Figure S12. Kyte-Doolittle hydropathy plot of the NET isoform P23975-3.

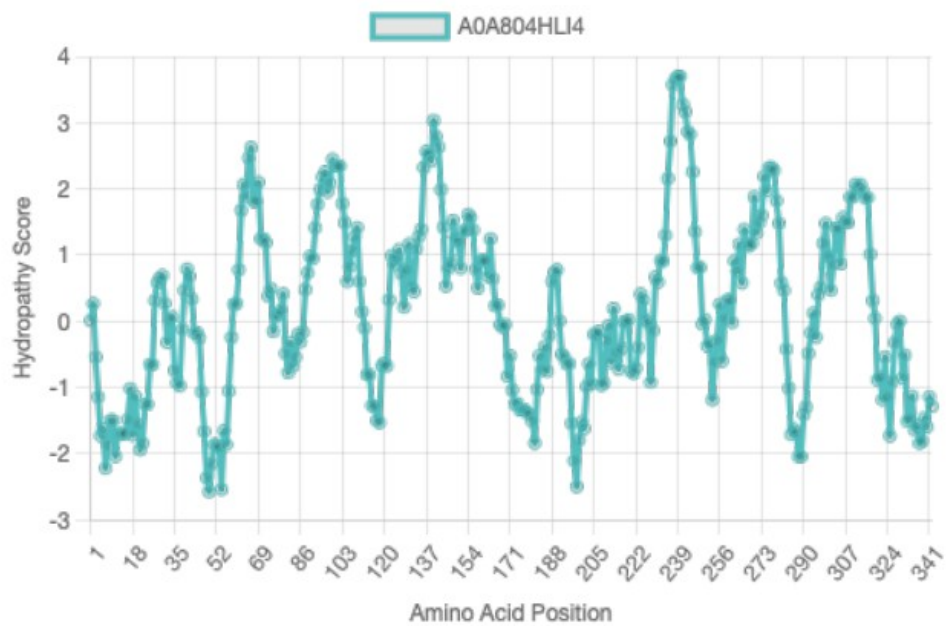


Figure S13. Kyte-Doolittle hydropathy plot of the NET isoform A0A804HLI4.

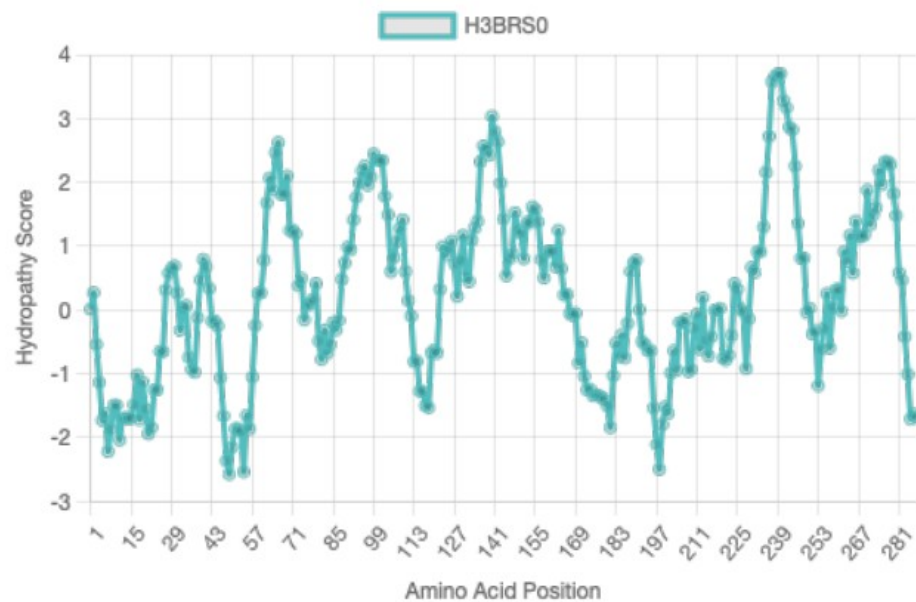


Figure S14. Kyte-Doolittle hydropathy plot of the NET isoform H3BRS0.

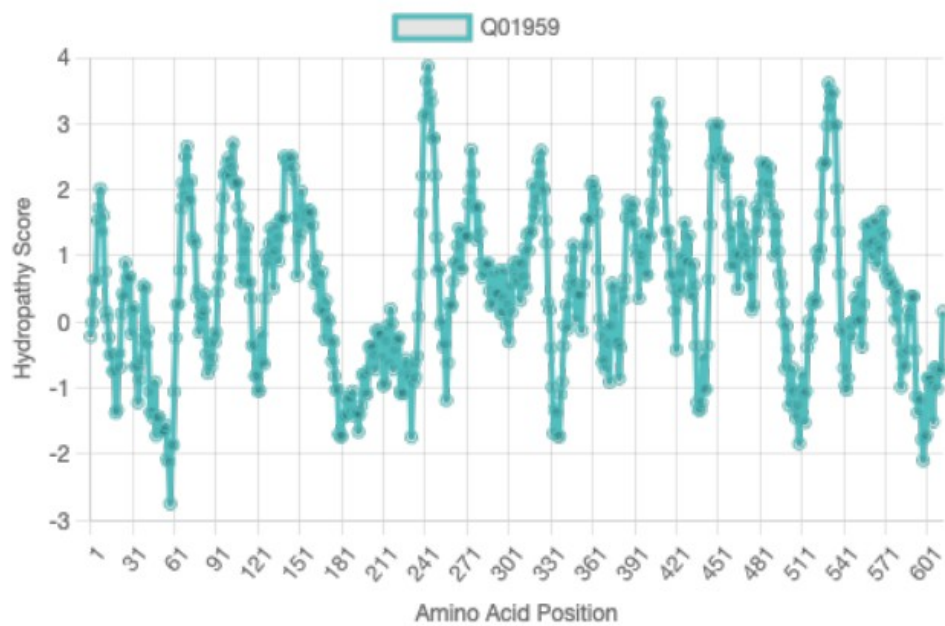


Figure S15. Kyte-Doolittle hydropathy plot of the canonical DAT isoform Q01959.

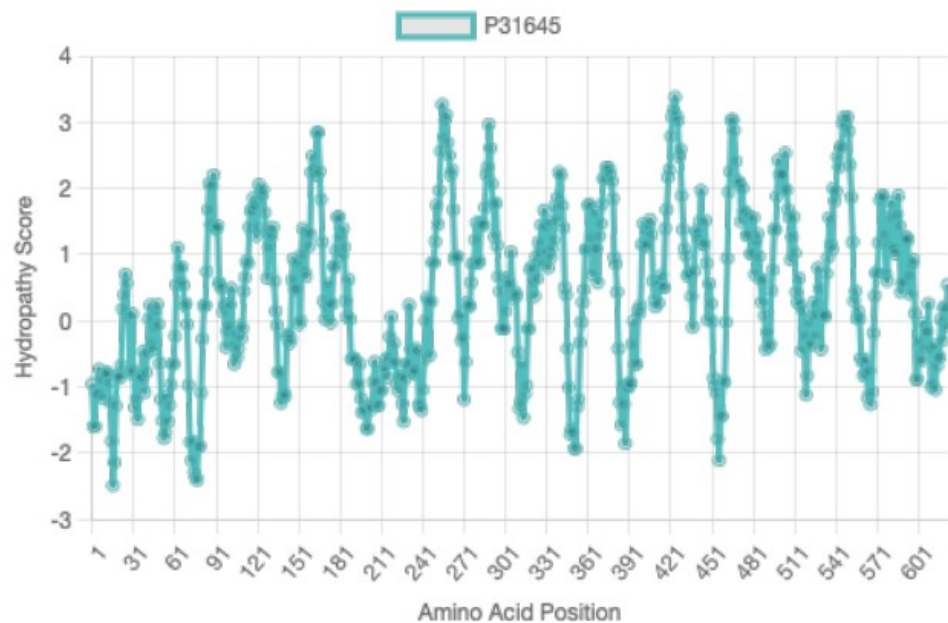


Figure S16. Kyte-Doolittle hydropathy plot of the canonical SERT isoform P31645.

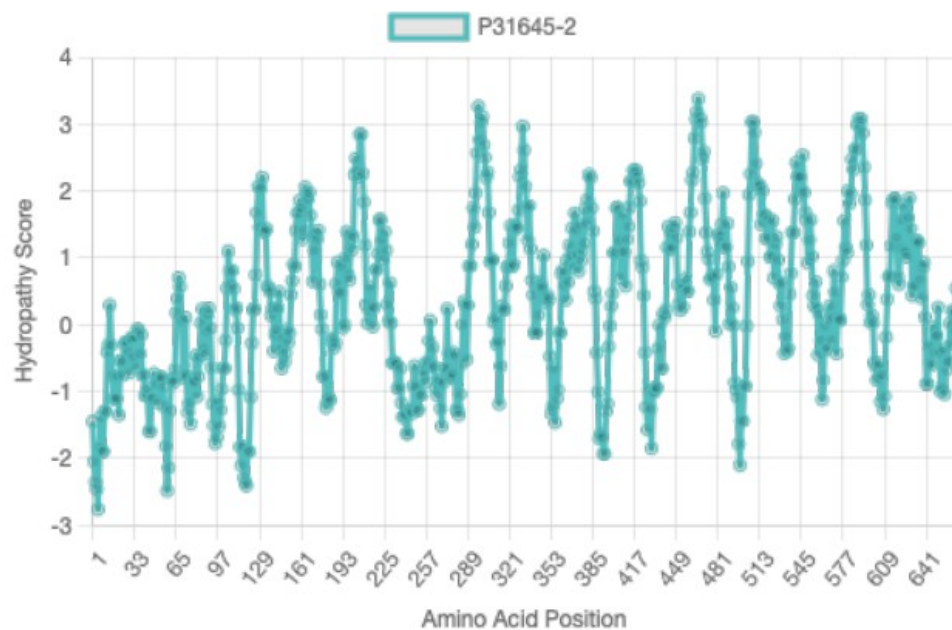


Figure S17. Kyte-Doolittle hydropathy plot of the SERT isoform P31645-2.

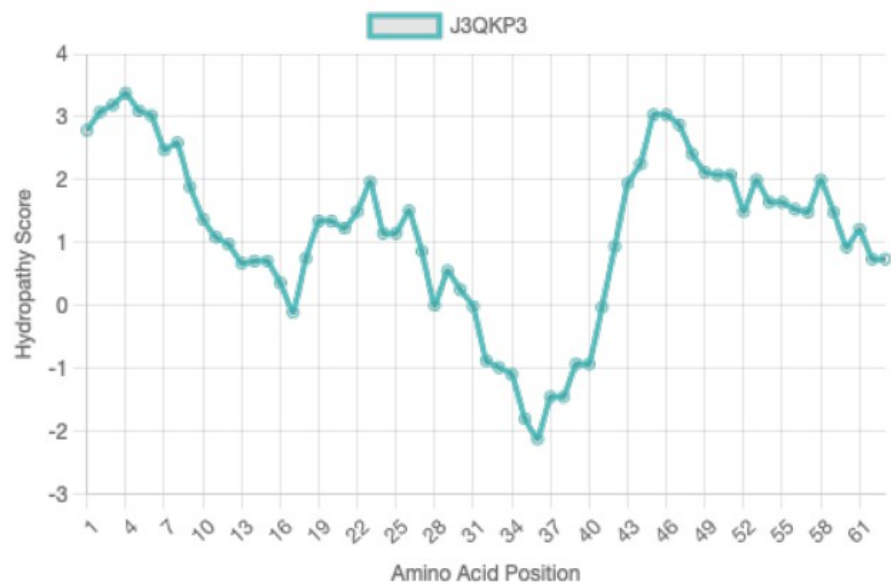


Figure S18. Kyte-Doolittle hydropathy plot of the SERT isoform J3QKP3.

Table S2. AlphaFold3 prediction confidence metrics for the monomeric and multi-chain (complex) models.

Model	Assembly	pTM	ipTM
NET (P23975)	Monomer	0.87	—
DAT (Q01959)	Monomer	0.86	—
SERT (P31645)	Monomer	0.83	—
A0A804HLI4 (isoform)	Monomer	0.61	—
NET–NET homodimer	Complex	0.55	0.28
DAT–DAT homodimer	Complex	0.54	0.27
SERT–SERT homodimer	Complex	0.49	0.16
A0A804HLI4– A0A804HLI4	Complex	0.22	0.16
NET–A0A804HLI4	Complex	0.54	0.13
DAT–A0A804HLI4	Complex	0.54	0.14
SERT–A0A804HLI4	Complex	0.58	0.18

Values are from the top-ranked model of five AlphaFold3 predictions per job. pTM, predicted template modeling score; ipTM, interface predicted template modeling score (reported for multi-chain assemblies only). The low complex ipTM values (0.13–0.28) arise because AlphaFold3 predicts transmembrane interfaces without an explicit lipid environment; this motivated the explicit lipid-bilayer molecular-dynamics relaxation of all complexes described in the Methods.